GraphSmile reproduction: paper vs clean-protocol runs

Branch experiments/p0-protocol. Every "ours" number is read from results/logs and results/experiments by scripts/build_comparison_pdf.py. Test set: IEMOCAP-6 (31 dialogues, 1623 utterances). Paper: Li et al., IEEE TPAMI 2025, Table IV.

1. Protocols compared

A. Original code, test-selected, 1 seed (IEMOCAP_full.log): epoch with the best TEST F1; optimistic. **B. p0**: 5 seeds, epoch by valid_f1_emo, but the repo validation dialogues are also training dialogues (found in pass 1), so the selector is a train-set metric. **C. p0c (baseline)**: 5 seeds, --clean_valid (108 train and 12 held-out valid dialogues, 0 overlap), epoch by valid_f1_emo. **D. p1c**: C plus --class_weight. The paper does not state how its reported epoch is chosen.

2. IEMOCAP-6, per-class F1 (%)

Class	Paper	A: orig, 1 seed	B: p0, 5 seeds	C: p0c clean, 5 seeds	D: p1c +cw, 5 seeds
Happy	63.09	54.87	58.69 ± 3.18	45.62 ± 4.17	44.80 ± 4.34
Sad	83.16	82.14	82.34 ± 0.67	79.89 ± 1.58	81.05 ± 1.24
Neutral	71.07	70.85	69.57 ± 0.23	67.98 ± 2.22	68.60 ± 0.99
Angry	71.38	68.92	68.19 ± 0.75	68.87 ± 1.56	68.20 ± 0.98
Excited	79.66	78.04	77.57 ± 0.92	76.10 ± 1.24	76.36 ± 1.11
Frustrated	66.84	66.22	65.12 ± 0.49	63.21 ± 2.47	62.73 ± 1.88
ACC	**72.77**	**71.23**	**70.82 ± 0.44**	**68.65 ± 1.21**	**68.87 ± 0.63**
WF1	**72.81**	**71.17**	**70.82 ± 0.47**	**68.26 ± 0.94**	**68.38 ± 0.62**

Mean ± sample std (ddof=1) over seeds 2024, 1, 2, 3, 4. Column A is a single seed.

3. Where the gap comes from (diagnostic only; never used to choose anything)

Mean WF1 over 5 seeds	p0	p0c	p1c
Selected on valid (reported)	70.82	68.26	68.38
Final epoch (120)	70.00	68.04	70.48
Mean of last 10 epochs	69.76	68.57	69.73
Best test epoch ("oracle", optimistic)	71.61	71.22	71.01
Paper WF1	72.81		

Under every protocol training reaches about 71 WF1 at its best test epoch, but the epoch picked on the 12-dialogue held-out validation set gives about 68.3 for both p0c and p1c. For p0c this is close to simply taking the last epoch (68.04); for p1c the final epoch is higher (70.48), a post-hoc observation from one set of 5 seeds that is not part of the pass-1 decision rule. Valid WF1 is flat across seeds while test moves about 3 points between epochs.

4. Happy / Excited confusion (test)

Run	Happy to Excited (of 144), per seed	mean	Excited to Happy (of 299), per seed	mean
p0 (overlapping valid)	48, 37, 32, 36, 38	38.2	33, 34, 42, 31, 32	34.4
p0c (clean baseline)	51, 46, 51, 83, 77	61.6	34, 38, 24, 16, 15	25.4
p1c (class weight)	66, 65, 54, 83, 65	66.6	24, 20, 51, 14, 19	25.6

Class weighting (pass 1) did not reduce the confusion. Decision: inconclusive; WF1 changed by +0.11 and Happy by -0.81 against baseline stds of 0.94 and 4.17.

5. Other datasets (unchanged since the first reproduction)

Dataset	Paper ACC / WF1	Ours ACC / WF1 (1 seed, test-selected)
IEMOCAP-4	86.53 / 86.52	86.00 / 85.98
MELD	67.70 / 66.71	67.59 / 66.55
CMU-MOSEI	46.82 / 44.93	45.10 / 43.96

Not rerun under the clean protocol or with several seeds, so they carry the same test-selection caveat as column A and are not comparable to sections 2 to 4.

6. Status and open items

The paper epoch-selection protocol is not stated in the text read (arXiv 2407.21536v2, Sections I to V-A). Pass 2 (last-10-epoch weight averaging, --avg_last) is planned in results/experiments/p2-plan.md and is not implemented or run. Baseline rows, MSAC rows marked with a double dagger and Figs. 4 to 7 of the paper remain unreproduced (see REPRODUCTION.md).